Code No: 135CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Define attenuation in the context of signal transmission. [2]
b) What is the purpose of frequency translation in communication systems? [3]
c) What are the drawbacks of BPSK? [2]
d) What are the noise effects in Pulse Code Modulation? [3]
e) What is Ethernet? [2]
f) What is the role of a network switch in a LAN? [3]
g) What is the primary function of a transponder in a satellite communication system? [2]
h) What are the core and cladding in a fiber-optic cable, and how do they contribute is an efficient light propagation? [3]
i) Which multiple access technique does GSM employ to handle multiple users in a cell? [2]
j) What does WiMAX stand for, and in which type of network is it commonly used? [3]

PART – B

(50 Marks)

2.a) Explain the need for modulation in communication systems and provide examples of modulation techniques used in modern technologies.
b) Analyze the significance of the electromagnetic spectrum in modern communication and its impact on different technologies. [5+5]

OR

3.a) Evaluate the significance of decibels as a unit for expressing signal strength and the advantages it offers in comparing signal levels in logarithmic scale.
b) Describe the concept of frequency translation and how it is applied in various communication systems. [5+5]

4.a) Discuss the principles of Analog Modulation using AM and FM techniques, and compare their advantages and disadvantages in practical communication systems.
b) Examine the demodulation schemes used to recover the original signal from modulated carriers in digital modulation techniques, highlighting the key differences between the processes. [5+5]

OR

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5.a) Design a comprehensive communication system that incorporates multiple modulation techniques to transmit and receive signals efficiently in different operational conditions.

b) With a neat sketch, explain the modulation and detection of Differential Phase shift Keying. [5+5]

6.a) Discuss the evolution of telephones and telephone systems from the early days of Alexander Graham Bell to modern digital communication systems.

b) Examine the principles of Internet Telephony (VoIP) and how it leverages the Internet to transmit voice data. [5+5]

OR

7.a) Explain the concept of paging systems and how they are used for wireless communication.

b) Discuss the working principle of Token Ring LAN access method and how it ensures fair access to the network resources among connected devices. [5+5]

8.a) Explain the various types of satellite orbits, including geostationary, polar, and elliptical orbits. Discuss the advantages and disadvantages of each orbit type for satellite communication systems.

b) Discuss the operation of an optical transmitter and how it converts electrical signals into light signals for transmission through fiber-optic cables. [5+5]

OR

9.a) Discuss the concept of trilateration and its role in determining user locations in GPS systems.

b) Examine the different types of WDM, such as Coarse Wavelength Division Multiplexing(CWDM) and Dense Wavelength Division Multiplexing (DWDM), and discuss their respective applications and limitations in optical communication networks. [5+5]

10.a) Describe the basic architecture of a cellular telephone system, including the components such as base stations, mobile switching centers, and cell sites.

b) Explain the concept of Wireless LAN (WLAN) and how it enables wireless connectivity in local area networks? [5+5]

OR

11.a) Explain the fundamental principles of the Global System for Mobile Communications (GSM) and how it employs Time Division Multiple Access (TDMA) to handle multiple users in a cell?

b) Elaborate the characteristics of Zigbee as a low-power, short-range wireless communication technology. [5+5]

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